PROJECT PROPOSAL

# Smart Workforce Optimization and Resource Sharing System for Multi-Organization Call Centers

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# Problem Statement

### Idle staff, then overload

Demand changes hour by hour, but staff numbers stay fixed

### High labour cost

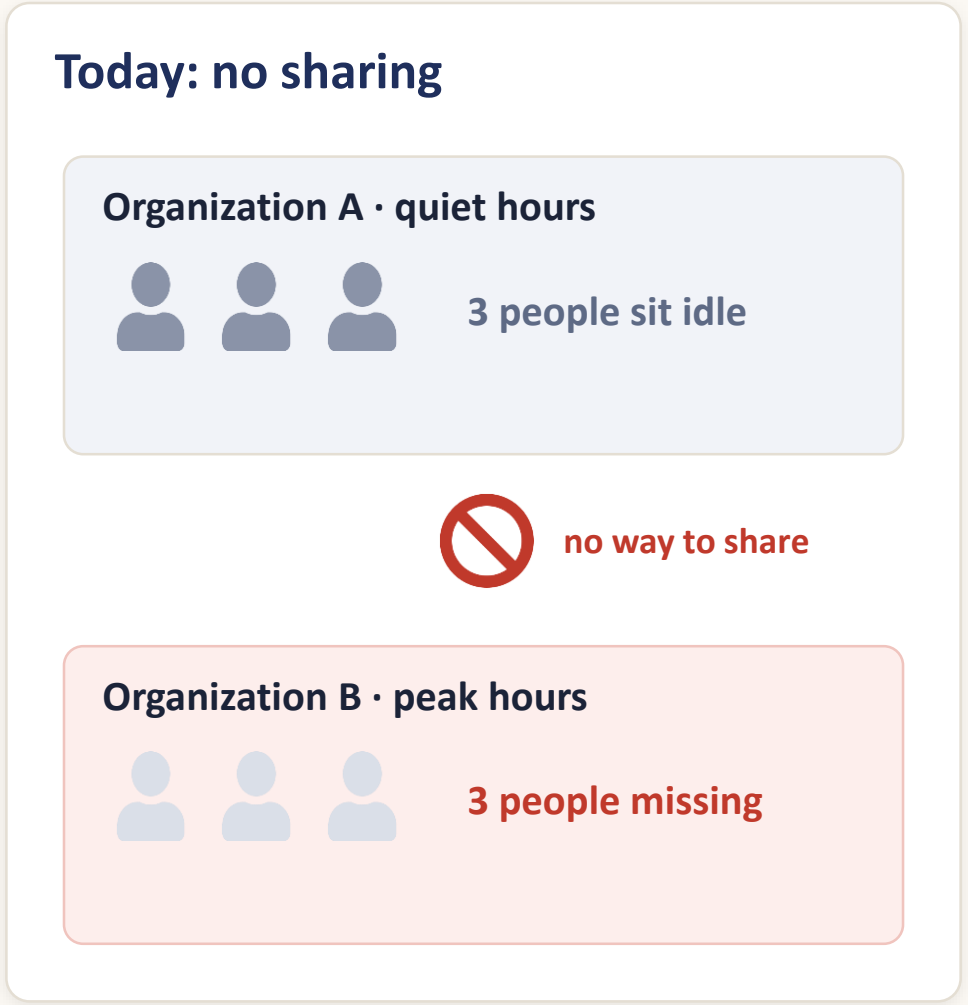
Each organization hires extra staff on its own for peak times

### Fixed teams only

Traditional systems cannot share people between organizations

### No live visibility

Nobody sees who is working where, right now



# Goals and Objectives

## Goals

- Optimize human resource utilization
- Reduce employee idle time
- Lower operational costs
- Maintain high service quality

## Objectives

- Build a hierarchical workforce system
- Manage fixed and on-demand employees
- Maintain one central shared employee pool
- Develop AI-based allocation, with employee consent
- Provide online dashboards for all users
- Enable VoIP calls and live monitoring

# Our Solution

Business Owner › Central Admin › Organization › Org. Manager › Employee

### Shared employee pool
One pool of trained people, shared by all organizations

### AI autopilot
Picks the best fit by skills, rating and availability

### Loans and recall
Spare staff are lent out and recalled within 3 minutes

### Employee consent
Employees accept or decline every assignment

### Live monitoring and calls
Real-time status, WebRTC audio and video calls

### Billing and reports
Invoices, online payments (Safepay), notifications

**What is new:** staff sharing between many organizations under an agreement, decided by AI, accepted by the employee. Existing tools serve one organization only.

Built with React · ASP.NET Core · SQL Server · Python (FastAPI) AI service · Socket.IO · WebRTC

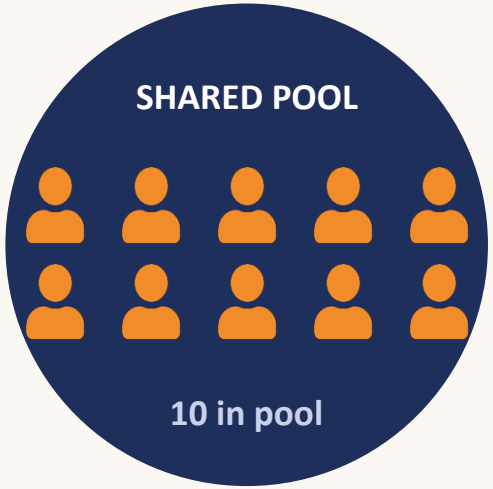
# How the shared pool works

8:30 AM

10 trained people are shared by 5 organizations. Each organization says what it needs.

**Bus Service**
needs 2

**University**
needs 2




**Restaurant**
needs 1

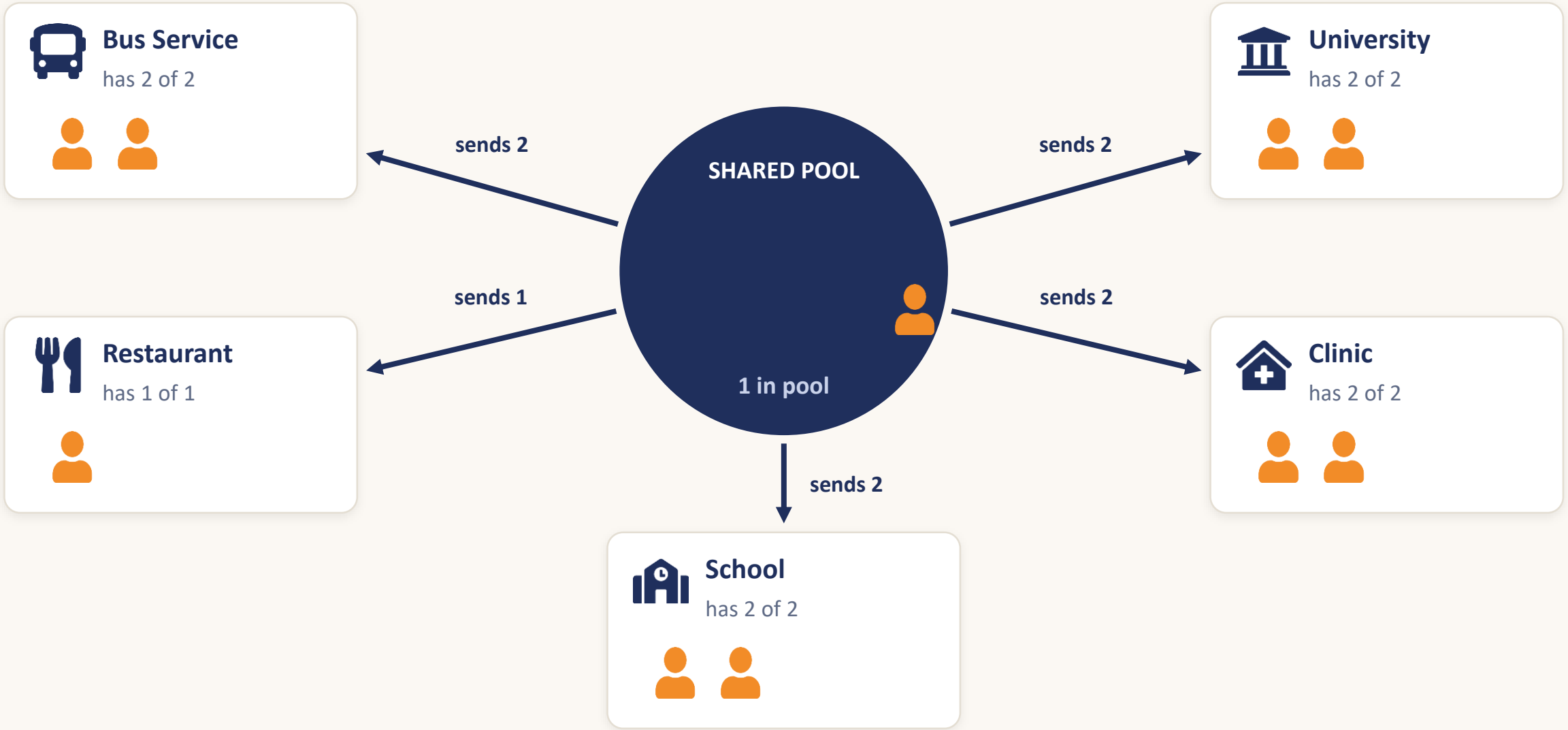
**Clinic**
needs 2

**School**
needs 2

# How the shared pool works

9 AM – 2 PM

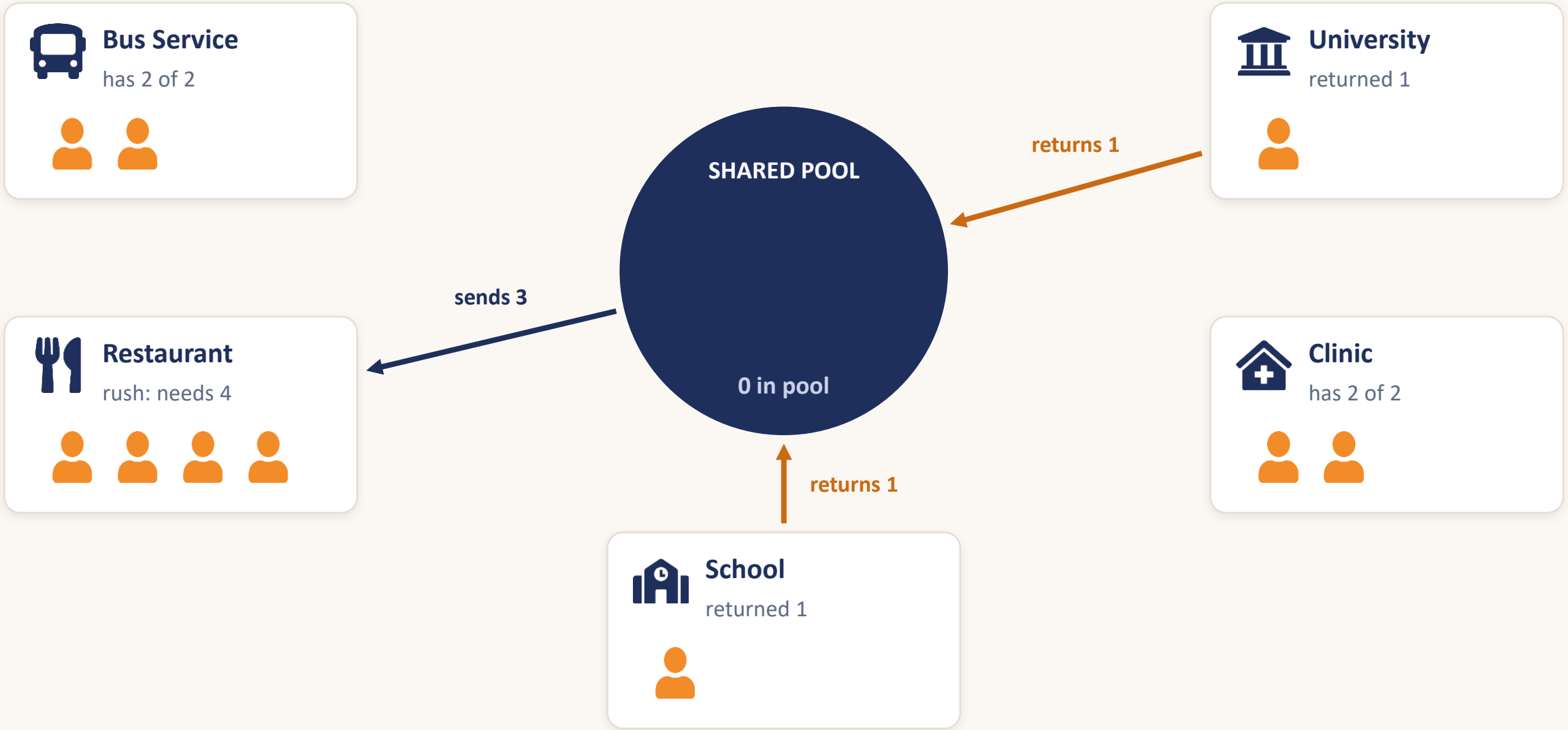
The AI autopilot sends the best-fit people: skills, rating and availability.



# How the shared pool works

2 PM

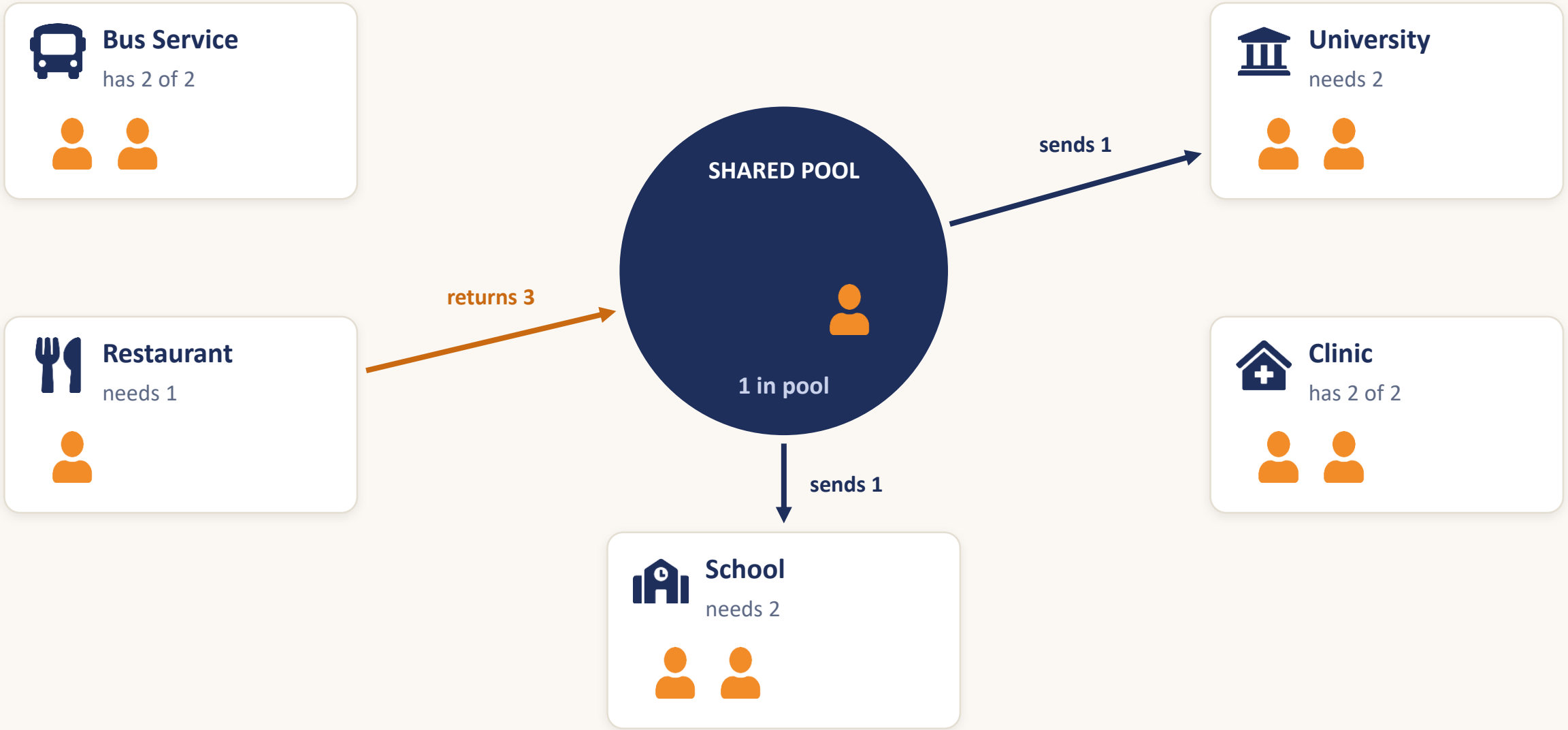
University and School return one person each. The restaurant rush gets all 3 free people.



# How the shared pool works

**3 PM – 5 PM**

The rush is over: 3 people go back and are re-assigned where they are needed now.



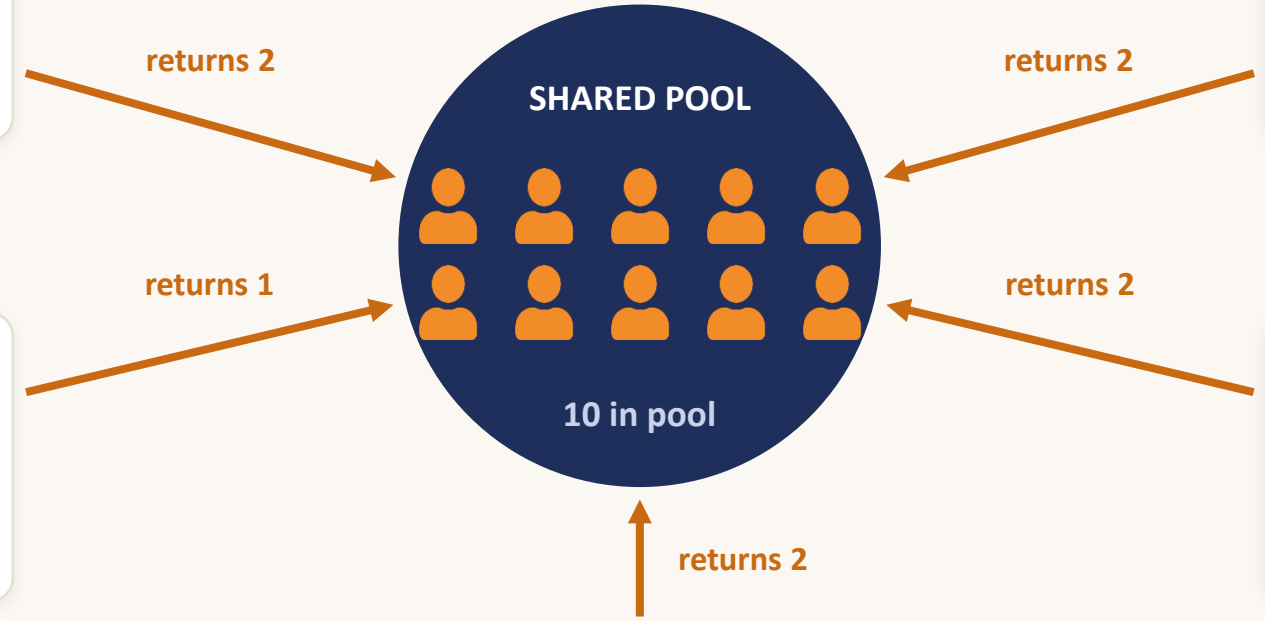
# How the shared pool works

**5 PM**

End of the day: everybody is back in the shared pool.

**Bus Service**
back in pool

**University**
back in pool




**Restaurant**
back in pool

**Clinic**
back in pool

**School**
back in pool

**10 people served 5 organizations all day: no extra hiring, no idle staff.**

- Less idle time
- Lower cost
- Better service